University Name: Babu Banarasi Das University University Rollno: 1250258153

Student Name: Avani Singh Course: BCA DS AI

Batch: BCA DS12

Faculty Name: Mr. Ankit Verma Project Name: IPL Data Analysis

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

matches = pd.read_csv('/content/dr
deliveries = pd.read_csv('/content
```

i) Match won by the maximum margin of runs

```
matches.loc[matches['win_by_runs']
```

	43
id	44
season	2017
city	Delhi
date	2017-05-06
team1	Mumbai Indians
team2	Delhi Daredevils
toss_winner	Delhi Daredevils
toss_decision	field
result	normal
dl_applied	0
winner	Mumbai Indians
win_by_runs	146
win_by_wickets	0
player_of_match	LMP Simmons
venue	Feroz Shah Kotla
umpire1	Nitin Menon
umpire2	CK Nandan
umpire3	NaN

dtype: object

Insight

This shows the match with the largest victory margin by runs.

It indicates a dominant performance by the winning team.

ii) Match won by maximum wickets

```
matches.loc[matches['win_by_wickets'].idxmax()]
```

	2
id	3
season	2017
city	Rajkot
date	2017-04-07
team1	Gujarat Lions
team2	Kolkata Knight Riders
toss_winner	Kolkata Knight Riders
toss_decision	field
result	normal
dl_applied	0
winner	Kolkata Knight Riders
win_by_runs	0
win_by_wickets	10
player_of_match	CA Lynn
venue	Saurashtra Cricket Association Stadium
umpire1	Nitin Menon
umpire2	CK Nandan
umpire3	NaN

dtype: object

Insight

This identifies the match where the team won with the highest wickets remaining.

It indicates an easy run chase with strong batting performance.

Such wins usually mean the opponent bowling attack struggled.

iii) Match won by the minimum margin of runs

```
matches[matches['win_by_runs']>0].loc[(matches[matches['win_l
```

	58
id	59
season	2017
city	Hyderabad
date	2017-05-21
team1	Mumbai Indians
team2	Rising Pune Supergiant
toss_winner	Mumbai Indians
toss_decision	bat
result	normal
dl_applied	0
winner	Mumbai Indians
win_by_runs	1
win_by_wickets	0
player_of_match	KH Pandya
venue	Rajiv Gandhi International Stadium, Uppal
umpire1	NJ Llong
umpire2	S Ravi
umpire3	NaN

dtype: object

Insight

This shows the closest run victory in IPL matches.

These matches are usually very competitive and thrilling.

Even a small difference in runs can decide the match result.

iv) Match won by minimum wickets

```
matches[matches['win_by_wickets']>0].loc[(matches[matches['w
```

	559
id	560
season	2015
city	Kolkata
date	2015-05-09
team1	Kings XI Punjab
team2	Kolkata Knight Riders
toss_winner	Kings XI Punjab
toss_decision	bat
result	normal
dl_applied	0
winner	Kolkata Knight Riders
win_by_runs	0
win_by_wickets	1
player_of_match	AD Russell
venue	Eden Gardens
umpire1	AK Chaudhary
umpire2	HDPK Dharmasena
umpire3	NaN

dtype: object

Insight

This shows matches where the chasing team barely won with very few wickets left.

Such matches show high pressure situations and exciting finishes.

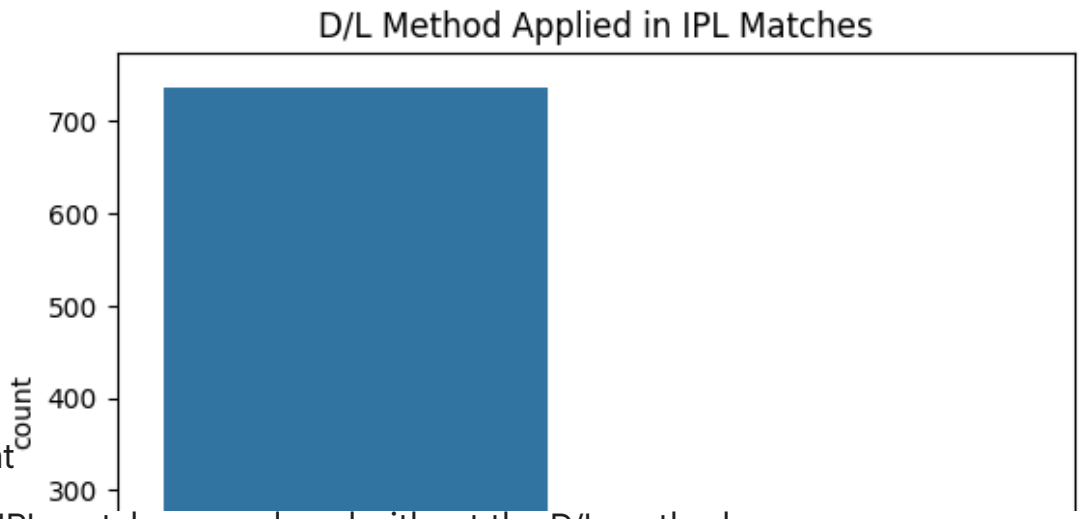
v) Matches where D/L method was and wasn't applied

```
matches['dl_applied'].value_counts()
```

dl_applied	count
0	737
1	19

dtype: int64

```
sns.countplot(x='dl_applied', data=matches)
plt.title("D/L Method Applied in IPL Matches")
plt.show()
```



Insight

Most IPL matches are played without the D/L method.

D/L method is used only when rain interrupts the match.

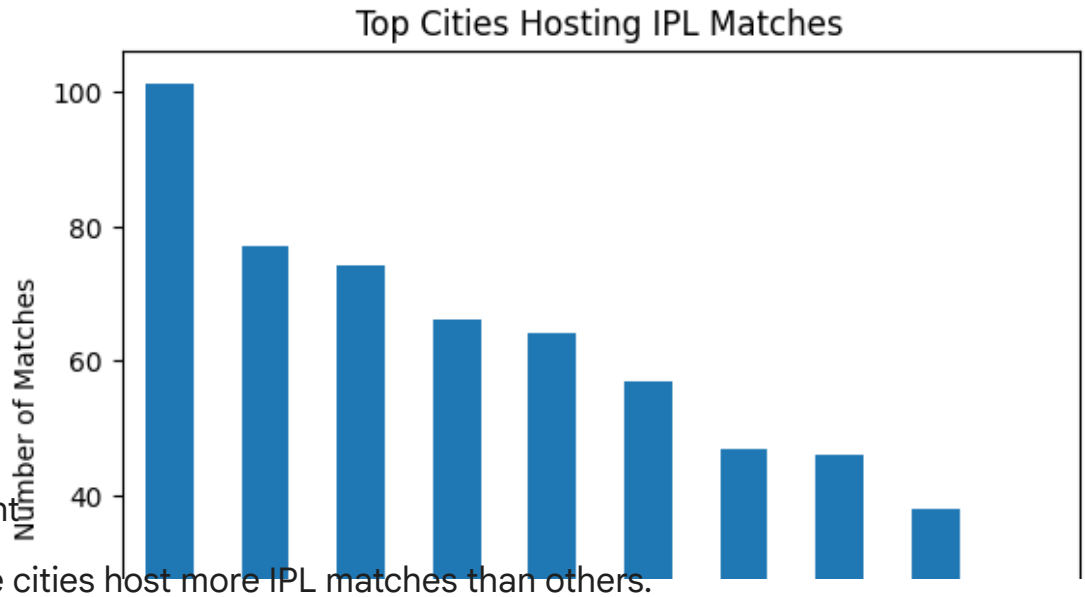
vi) Number of matches held in each city

```
matches['city'].value_counts()
```

city	count
Mumbai	101
Kolkata	77
Delhi	74
Bangalore	66
Hyderabad	64
Chennai	57
Jaipur	47
Chandigarh	46
Pune	38
Durban	15
Bengaluru	14
Visakhapatnam	13
Centurion	12
Ahmedabad	12
Mohali	10
Rajkot	10
Dharamsala	9
Indore	9
Johannesburg	8
Port Elizabeth	7
Abu Dhabi	7
Cape Town	7
Ranchi	7
Cuttack	7
Sharjah	6

Raipur	6
Kochi	5
Kanpur	4
Kimberley	3
Nagpur	3

```
matches['city'].value_counts().head(10).plot(kind='bar')
plt.title("Top Cities Hosting IPL Matches")
plt.xlabel("City")
plt.ylabel("Number of Matches")
plt.show()
```



Insight

Some cities host more IPL matches than others.

Cities with popular teams and large stadiums usually host more matches.

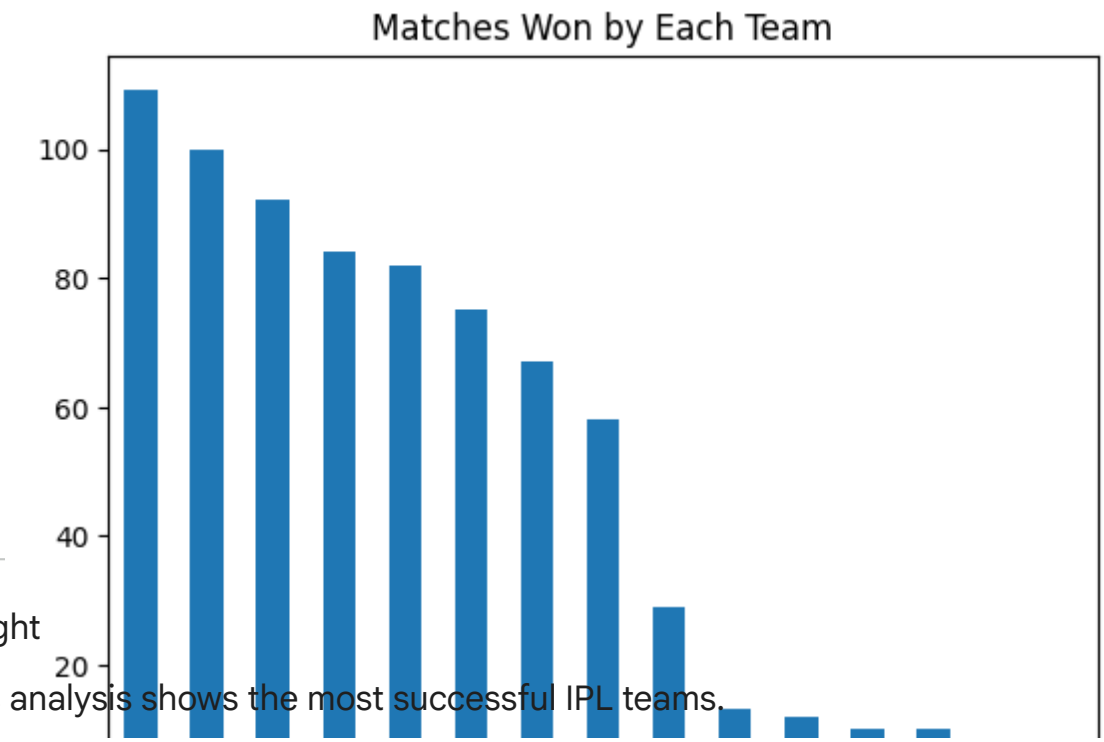
vii) Number of matches won by each team

```
matches['winner'].value_counts()
```

winner	count
Mumbai Indians	109
Chennai Super Kings	100
Kolkata Knight Riders	92
Royal Challengers Bangalore	84
Kings XI Punjab	82
Rajasthan Royals	75
Delhi Daredevils	67
Sunrisers Hyderabad	58
Deccan Chargers	29
Gujarat Lions	13
Pune Warriors	12
Rising Pune Supergiant	10
Delhi Capitals	10
Kochi Tuskers Kerala	6
Rising Pune Supergiants	5

dtype: int64

```
matches['winner'].value_counts().plot(kind='bar')
plt.title("Matches Won by Each Team")
plt.show()
```



Insight

This analysis shows the most successful IPL teams.

Teams with more wins generally have strong players and better strategies.

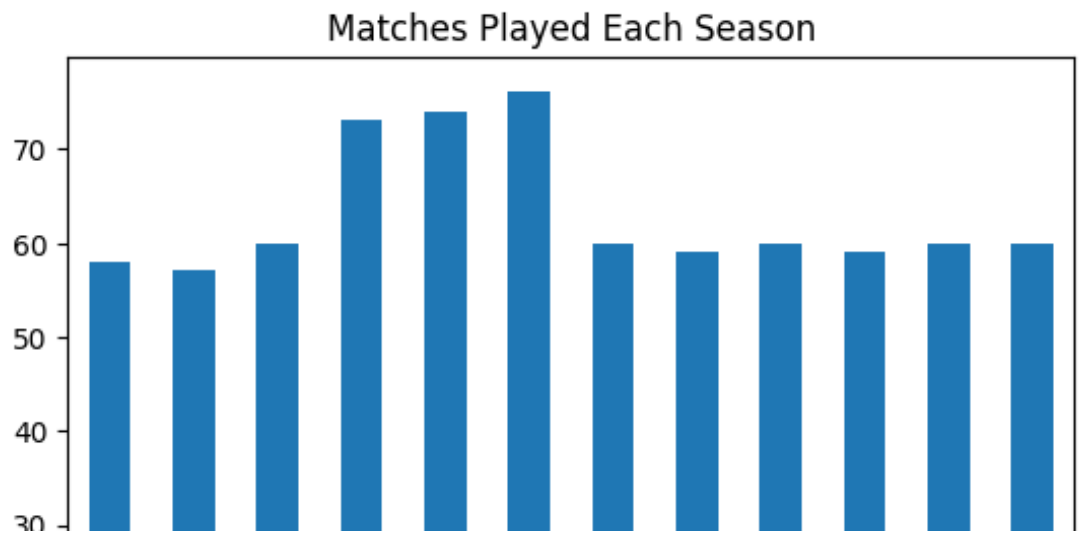
viii) Number of matches held every season

```
matches['season'].value_counts().sort_index()
```

season	count
2008	58
2009	57
2010	60
2011	73
2012	74
2013	76
2014	60
2015	59
2016	60
2017	59
2018	60
2019	60

dtype: int64

```
matches['season'].value_counts().sort_index().plot(kind='bar
plt.title("Matches Played Each Season")
plt.show()
```



Insight

The number of matches varies each season.

Some seasons had more teams and more matches.

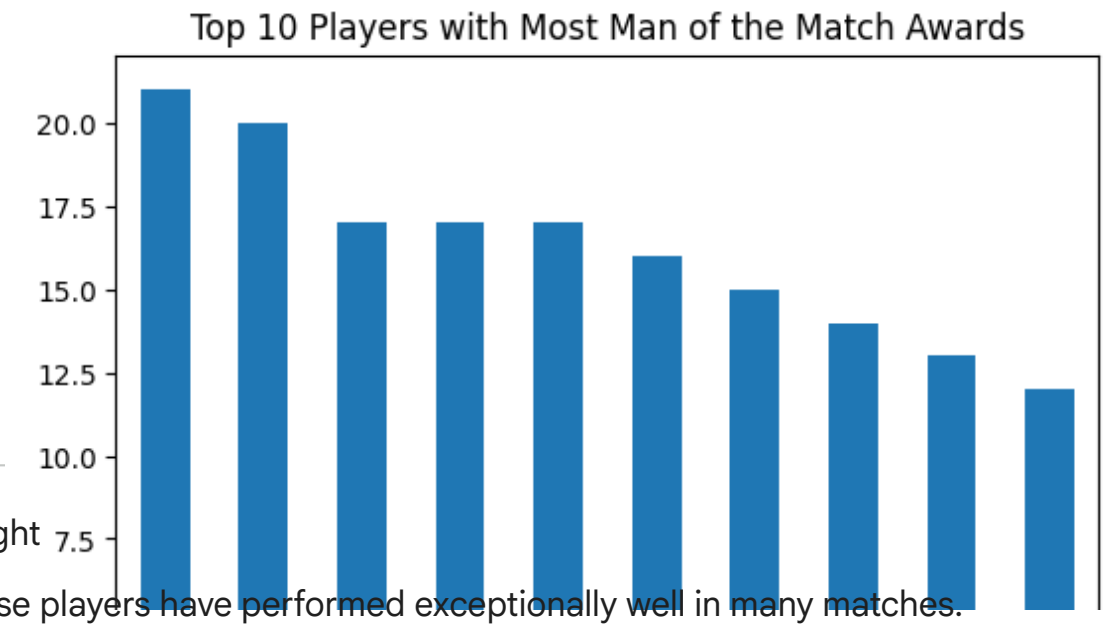
ix) Top 10 players based on Man of the Match awards

```
matches['player_of_match'].value_counts().head(10)
```

player_of_match	count
CH Gayle	21
AB de Villiers	20
MS Dhoni	17
RG Sharma	17
DA Warner	17
YK Pathan	16
SR Watson	15
SK Raina	14
G Gambhir	13
MEK Hussey	12

dtype: int64

```
matches['player_of_match'].value_counts().head(10).plot(kind=
plt.title("Top 10 Players with Most Man of the Match Awards"
plt.show()
```



Insight

These players have performed exceptionally well in many matches.

Frequent awards indicate consistent match-winning performances.

x) Does winning the toss mean winning the match?

```
toss_win = matches[matches['toss_winner'] == matches['winner
print("Matches where toss winner also won match:", toss_win.s
print("Total matches:", matches.shape[0])
```

```
Matches where toss winner also won match: 393
Total matches: 756
```

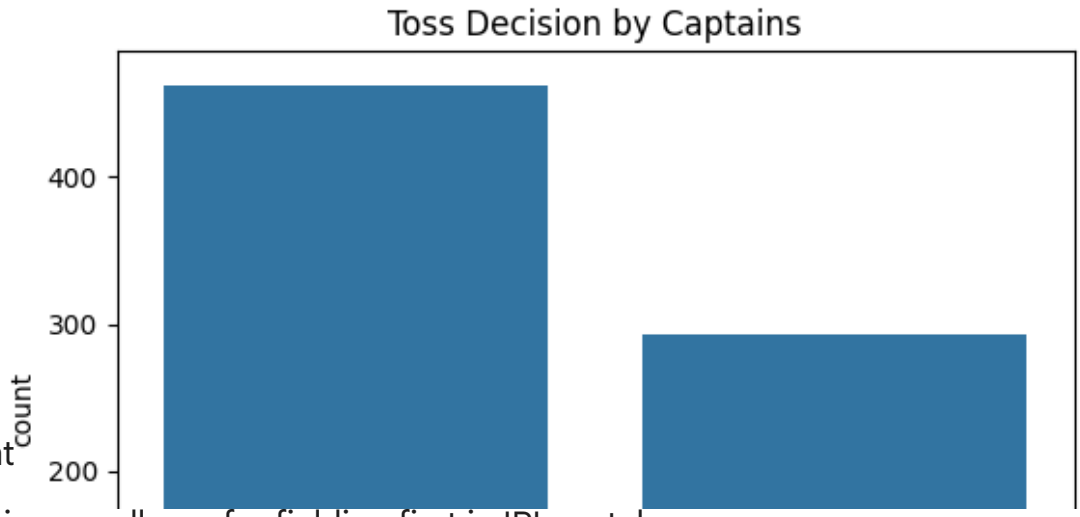
xi) Decision taken by captains after winning the toss

```
matches['toss_decision'].value_counts()
```

toss_decision	count
field	463
bat	293

dtype: int64

```
sns.countplot(x='toss_decision', data=matches)
plt.title("Toss Decision by Captains")
plt.show()
```



Insight

Captains usually prefer fielding first in IPL matches.

Chasing targets is often considered easier due to pitch conditions.

xii) Matches where D/L method applied every season

```
matches.groupby('season')['dl_applied'].sum()
```

season	dl_applied
2008	2
2009	3
2010	0
2011	3
2012	0
2013	0
2014	1
2015	2
2016	4
2017	1
2018	3
2019	0

dtype: int64

Insight

Some seasons had rain interruptions causing D/L method usage.

However, it occurs in very few matches overall.

xiii) Different results for matches

```
matches['result'].value_counts()
```

result	count
normal	743
tie	9
no result	4

dtype: int64

Insight

Most IPL matches end with normal results.

A few matches may result in ties or no results.

xiv) How many times a team won the toss

```
matches['toss_winner'].value_counts()
```

toss_winner	count
Mumbai Indians	98
Kolkata Knight Riders	92
Chennai Super Kings	89
Kings XI Punjab	81
Royal Challengers Bangalore	81
Delhi Daredevils	80
Rajasthan Royals	80
Sunrisers Hyderabad	46
Deccan Chargers	43
Pune Warriors	20
Gujarat Lions	15
Delhi Capitals	10
Kochi Tuskers Kerala	8
Rising Pune Supergiants	7
Rising Pune Supergiant	6

dtype: int64

Insight

Some teams win the toss more frequently due to probability over many matches.

xv) Best venue for defending and chasing a total

```
matches.groupby('venue')['win_by_runs'].mean().sort_values(a
```

venue	win_by_runs
Himachal Pradesh Cricket Association Stadium	31.000000
Rajiv Gandhi Intl. Cricket Stadium	30.375000
Buffalo Park	29.000000
St George's Park	28.571429
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium	25.636364

dtype: float64

```
matches.groupby('venue')['win_by_wickets'].mean().sort_values
```

venue	win_by_wickets
Holkar Cricket Stadium	6.444444
Green Park	5.500000
IS Bindra Stadium	5.000000
Saurashtra Cricket Association Stadium	4.800000
New Wanderers Stadium	4.375000

dtype: float64

Insight

Some stadiums favor defending a total due to pitch conditions.

Others are better for run chasing.

xvi) Best defending and chasing teams

```
matches.groupby('winner')['win_by_runs'].mean().sort_values(a
```

winner	win_by_runs
Chennai Super Kings	17.780000
Mumbai Indians	17.119266
Royal Challengers Bangalore	14.904762
Deccan Chargers	14.517241
Rising Pune Supergiant	12.300000
Sunrisers Hyderabad	12.293103
Rajasthan Royals	11.933333
Kolkata Knight Riders	11.804348
Pune Warriors	11.583333
Kings XI Punjab	11.280488
Rising Pune Supergiants	10.600000
Delhi Daredevils	10.074627
Delhi Capitals	9.200000
Kochi Tuskers Kerala	3.833333
Gujarat Lions	0.076923

dtype: float64

```
matches.groupby('winner')['win_by_wickets'].mean().sort_valu
```

winner	win_by_wickets
Gujarat Lions	5.000000
Kochi Tuskers Kerala	5.000000
Delhi Daredevils	4.104478
Rising Pune Supergiants	4.000000
Kolkata Knight Riders	3.815217
Royal Challengers Bangalore	3.738095
Rajasthan Royals	3.640000
Kings XI Punjab	3.268293
Rising Pune Supergiant	3.100000
Pune Warriors	3.000000
Sunrisers Hyderabad	2.982759
Delhi Capitals	2.900000
Chennai Super Kings	2.890000
Mumbai Indians	2.816514
Deccan Chargers	2.482759

dtype: float64

Insight

Some teams are stronger in defending targets due to good bowling.

Other teams perform better in chasing runs because of strong batting

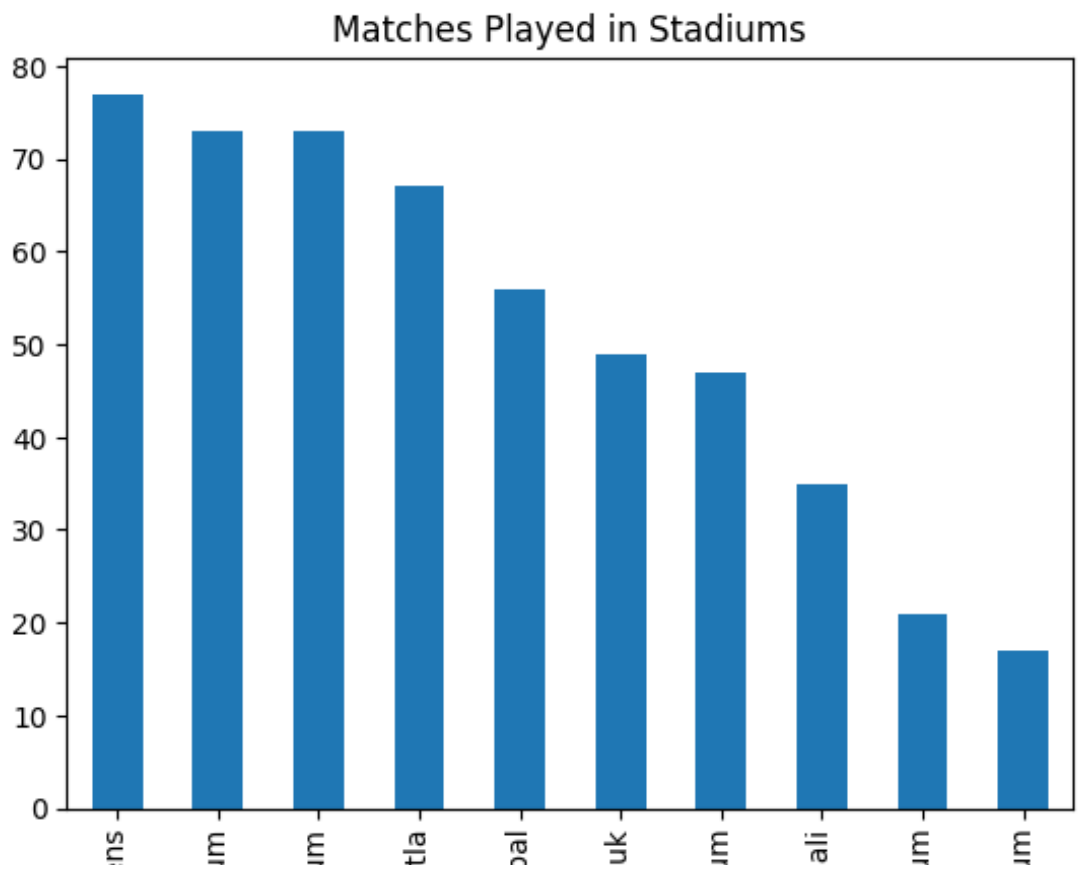
xvii) Number of matches played in different stadiums

```
matches['venue'].value_counts()
```

venue	count
Eden Gardens	77
Wankhede Stadium	73
M Chinnaswamy Stadium	73
Feroz Shah Kotla	67
Rajiv Gandhi International Stadium, Uppal	56
MA Chidambaram Stadium, Chepauk	49
Sawai Mansingh Stadium	47
Punjab Cricket Association Stadium, Mohali	35
Maharashtra Cricket Association Stadium	21
Subrata Roy Sahara Stadium	17
Dr DY Patil Sports Academy	17
Kingsmead	15
Punjab Cricket Association IS Bindra Stadium, Mohali	14
Sardar Patel Stadium, Motera	12
SuperSport Park	12
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium	11
Brabourne Stadium	11
Saurashtra Cricket Association Stadium	10

Holkar Cricket Stadium	9
Himachal Pradesh Cricket Association Stadium	9
New Wanderers Stadium	8
M. A. Chidambaram Stadium	8
Rajiv Gandhi Intl. Cricket Stadium	8
Newlands	7
Barabati Stadium	7
IS Bindra Stadium	7
M. Chinnaswamy Stadium	7
Dubai International Cricket Stadium	7
JSCA International Stadium Complex	7
St George's Park	7

```
matches['venue'].value_counts().head(10).plot(kind='bar')
plt.title("Matches Played in Stadiums")
plt.show()
```



Insight

Certain stadiums host more IPL matches due to team popularity and infrastructure.

These venues often have large fan attendance and better facilities.

Conclusion

This IPL data analysis project helped understand team performance, player achievements, and match trends across different seasons. The analysis showed how factors like toss decisions, venues, and player performances influence match results. Overall, the project demonstrates how data analysis can provide useful insights about IPL matches and team strategies.